

AUN ALI

HEALTHCARE SOLUTION ARCHITECT | ENTERPRISE INTEGRATION | PUBLIC HEALTH TECHNOLOGY

Katy, Texas | (832) 235-5025 | aunaliadenwala@hotmail.com | LinkedIn | Portfolio | GitHub

PROFESSIONAL SUMMARY

Healthcare Solution Architect and IT leader with 25+ years of experience designing, modernizing, integrating, and supporting enterprise healthcare and public health platforms. Deep hands-on expertise across HL7, FHIR, CDA/eICR, ELR, EDI X12, Mirth Connect, Rhapsody, SQL Server, .NET, Azure, Snowflake, Power BI, and API-based integration. Proven record leading mission-critical surveillance systems, international HIS implementations, data-center/cloud migrations, large-scale application modernization, vendor programs, and cross-functional technical teams.

ARCHITECTURE & LEADERSHIP HIGHLIGHTS

SELECTED IMPACT

- Lead a highly configurable public health disease surveillance platform serving 1,000+ users across multiple programs and disease workflows.
- Oversee an ESSENCE syndromic surveillance environment receiving approximately 300,000 HL7 messages daily from regional healthcare facilities.
- Rebuilt approximately 22 hospital information-system modules using Microsoft Windows/.NET technologies, including ADT, RIS, and CSSD/CICS workflows.
- Led or supported healthcare implementations across the United States, Pakistan, Afghanistan, and Kenya, including user training, infrastructure readiness, application configuration, and go-live support.
- Implemented EDI X12 warehouse-management integrations for Willis-Knighton Health System, including 850 Purchase Orders, 810 Invoices, 856 Ship Notices, and UB-04 to X12 837 claim conversion.
- Supported Point-of-Use and healthcare supply-chain solutions across 20 U.S. Department of Veterans Affairs hospital sites.

PROFESSIONAL EXPERIENCE

Houston, Texas

- Lead architecture, modernization, implementation, and production support for mission-critical public health applications, integration platforms, reporting solutions, and data-exchange initiatives.
- Provide technical leadership for the Health Department Surveillance System, a configurable platform supporting 1,000+ users, disease-specific workflows, questionnaires, investigations, and reporting.
- Lead healthcare interoperability solutions for HL7, ELR, eICR/CDA, ADT, immunization, laboratory, and public health data exchange using Mirth Connect, Rhapsody, SQL Server, .NET, APIs, XML, and JSON.
- Serve as Product Owner for Laboratory Information System modernization, partnering with laboratory stakeholders on requirements, RFP development, solution evaluation, and replacement planning.
- Implemented TB PACS modernization and migration from a legacy OPAL environment to a cloud-based radiology platform; provide technical leadership for TB case-management modernization and data migration.
- Lead ESSENCE infrastructure modernization and production operations for a syndromic surveillance platform receiving approximately 300,000 HL7 messages daily; coordinate current platform upgrade with the vendor.
- Architect and develop eICR processing capabilities for retrieval, CDA/XML transformation, validation, search, presentation, configurable mapping, and public health workflows.
- Led data-center modernization efforts, including application rebuilds, physical-to-virtual migrations, database migrations, interface-engine migrations, configuration validation, and production cutover.
- Design and support enterprise analytics and data modernization using SQL Server, Azure Data Factory, Snowflake, dimensional modeling, and Power BI, including immunization reporting and data-quality validation.
- Evaluate and implement AI-enabled solutions such as knowledge assistants/RAG, document processing, workflow automation, and configurable public health alerting while coordinating vendors, security, infrastructure, and program leadership.

Dallas, Texas

- Designed and implemented healthcare supply-chain and interoperability solutions using EDI X12, HL7, Mirth Connect, SQL Server, C#, .NET, WPF, and REST/SOAP services.
- Implemented EDI X12 warehouse-management integration for Willis-Knighton Health System, including 850 Purchase Orders, 810 Invoices, 856 Ship Notices/Manifests, and an EDI data mart for operational reporting.
- Designed and implemented UB-04 institutional claim transformation to ANSI X12 837 electronic healthcare claims.
- Automated receiving and point-of-care workflows and enhanced handheld/UWP applications supporting warehouse and inventory operations.
- Developed .NET/WPF applications including ZPL barcode/label printing and HL7-based integration between reporting systems and healthcare vendors; participated in FHIR-based EHR integration design.

Houston, Texas

- Developed and supported U.S. Department of Veterans Affairs Point-of-Use healthcare supply-chain solutions and implementations across 20 VA hospital sites.
- Designed HL7 integrations for VA VISN 10 RTLS and healthcare asset-management workflows using Mirth Connect, C#, nHapi, SQL Server, and WPF.
- Created a .NET RPC Broker and Windows/Web Service components integrating VA Vista/IFCAP/GIP legacy systems with modern supply-chain applications.
- Integrated supply-chain systems with Omnicell, Pyxis, Par Excellence, and other smart-cabinet platforms; automated RFID/LogiTag inventory workflows using SQL stored procedures and the Catamaran framework.

